

ISC Class XII Physics

Chapter 11

Dual Nature of Radiation and Matter

38 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Energy and Momentum of Photon, Work Function, Threshold Wavelength
- II. Einstein's Photoelectric Equation
- III. Stopping Potential
- IV. de Broglie Wavelength When Momentum or Velocity is Given
- V. de Broglie Wavelength When Kinetic Energy is Given
- VI. de Broglie Wavelength When Accelerating Potential is Given

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Formula Sheet

Constants used in this chapter:

$$h = 6.6 \times 10^{-34} \text{ J s} \quad c = 3 \times 10^8 \text{ m/s} \quad e = 1.6 \times 10^{-19} \text{ C}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg} \quad m_p = m_n = 1.67 \times 10^{-27} \text{ kg} \quad 1 \text{ \AA} = 10^{-10} \text{ m}$$

I. Photon Energy, Momentum, Work Function

Photon Energy

$$E = h\nu = \frac{hc}{\lambda}$$

ν = frequency, λ = wavelength

Photon Momentum

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

momentum carried by a photon

Dynamic Mass of Photon

$$m = \frac{E}{c^2}$$

rest mass of a photon is always zero

Threshold Frequency

$$\nu_0 = \frac{W_0}{h}$$

W_0 = work function of the metal

Threshold Wavelength

$$\lambda_0 = \frac{hc}{W_0}$$

longest wavelength that can cause emission

Photon Count from Power

$$n = \frac{P}{E_{\text{photon}}}$$

P = power of source, n = photons/s

II. Einstein's Photoelectric Equation

Einstein's Equation

$$E_{k(\max)} = h\nu - W_0 = \frac{hc}{\lambda} - W_0$$

$E_{k(\max)}$ = max. KE of ejected electron

Max. Speed of Photoelectron

$$v_{\max} = \sqrt{\frac{2E_{k(\max)}}{m_e}}$$

from max. kinetic energy

III. Stopping Potential

Stopping Potential Relation

$$eV_0 = h\nu - W_0$$

V_0 = stopping potential (volt)

Slope of V_0 - ν Graph

$$\text{slope} = \frac{h}{e}$$

gives Planck's constant from graph

Intercepts of V_0 - ν Graph

$$\nu_0 = \frac{W_0}{h}, \quad V_0|_{\nu=0} = -\frac{W_0}{e}$$

ν_0 = x-intercept, $-W_0/e$ = y-intercept

IV, V & VI. de Broglie Wavelength

de Broglie Wavelength (general)

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

p = momentum of the particle

de Broglie Wavelength (KE given)

$$\lambda = \frac{h}{\sqrt{2mE_k}}$$

E_k = kinetic energy of the particle

de Broglie Wavelength (electron, accelerating p.d. V)

$$\lambda = \frac{h}{\sqrt{2m_e eV}} \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$$

V in volts, standard result for electrons

I. Energy and Momentum of Photon, Work Function, Threshold Wavelength

Question 1

Find the frequency of a photon whose energy is 75 eV.

Given: $E = 75 \text{ eV} = 75 \times 1.6 \times 10^{-19} \text{ J} = 1.2 \times 10^{-17} \text{ J}$

Formula: $E = h\nu \Rightarrow \nu = \frac{E}{h}$

Solution:

$$\nu = \frac{1.2 \times 10^{-17}}{6.6 \times 10^{-34}} = 1.818 \times 10^{16} \text{ Hz}$$

Final Answer

$$\nu \approx 1.82 \times 10^{16} \text{ Hz}$$

Question 2

A beam of ultraviolet light has frequency $1.0 \times 10^{15} \text{ Hz}$. Find the energy of one quantum of this light, in eV. (ISC 2004)

★ Likely Exam Question

Given: $\nu = 1.0 \times 10^{15} \text{ Hz}$

Formula: $E = h\nu$

Solution:

$$E = 6.6 \times 10^{-34} \times 1.0 \times 10^{15} = 6.6 \times 10^{-19} \text{ J}$$

$$E = \frac{6.6 \times 10^{-19}}{1.6 \times 10^{-19}} = 4.125 \text{ eV}$$

Final Answer

$E \approx 4.13 \text{ eV}$ (often quoted as 4.14 eV using $h = 6.63 \times 10^{-34} \text{ Js}$)

Question 3

A spectral line has wavelength 400 nm. Calculate the frequency and the energy of the photons associated with it.

Given: $\lambda = 400 \text{ nm} = 4 \times 10^{-7} \text{ m}$

Formula: $\nu = \frac{c}{\lambda}, \quad E = h\nu$

Solution:

$$\nu = \frac{3 \times 10^8}{4 \times 10^{-7}} = 7.5 \times 10^{14} \text{ Hz}$$

$$E = 6.6 \times 10^{-34} \times 7.5 \times 10^{14} = 4.95 \times 10^{-19} \text{ J} = \frac{4.95 \times 10^{-19}}{1.6 \times 10^{-19}} = 3.09 \text{ eV}$$

Final Answer

$$\nu = 7.5 \times 10^{14} \text{ Hz}, \quad E \approx 4.95 \times 10^{-19} \text{ J} \approx 3.1 \text{ eV}$$

Common Mistake: Always convert nm to metres before substituting into $\nu = c/\lambda$.

Question 4

Calculate the number of photons emitted per second by a 40 W monochromatic source of light of wavelength 500 nm.

Given: $P = 40 \text{ W}, \quad \lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$

Formula: $E_{\text{photon}} = \frac{hc}{\lambda}, \quad n = \frac{P}{E_{\text{photon}}}$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5 \times 10^{-7}} = 3.96 \times 10^{-19} \text{ J}$$

$$n = \frac{40}{3.96 \times 10^{-19}} = 1.01 \times 10^{20} \text{ photons/s}$$

Final Answer

$$n \approx 1.0 \times 10^{20} \text{ photons per second}$$

Question 5

The energy of a photon is 42 eV. Find its momentum.

Given: $E = 42 \text{ eV} = 42 \times 1.6 \times 10^{-19} = 6.72 \times 10^{-18} \text{ J}$

Formula: $p = \frac{E}{c}$

Solution:

$$p = \frac{6.72 \times 10^{-18}}{3 \times 10^8} = 2.24 \times 10^{-26} \text{ kg m/s}$$

Final Answer

$$p = 2.24 \times 10^{-26} \text{ kg m/s}$$

Question 6

The energy of a photon is 10 eV. Calculate (i) its momentum, (ii) its wavelength, (iii) its rest mass and dynamic mass.

★ Likely Exam Question

Given: $E = 10 \text{ eV} = 1.6 \times 10^{-18} \text{ J}$

Formula: $p = \frac{E}{c}, \quad \lambda = \frac{h}{p}, \quad m = \frac{E}{c^2}$

Solution:

(i)

$$p = \frac{1.6 \times 10^{-18}}{3 \times 10^8} = 5.33 \times 10^{-27} \text{ kg m/s}$$

(ii)

$$\lambda = \frac{h}{p} = \frac{6.6 \times 10^{-34}}{5.33 \times 10^{-27}} = 1.2375 \times 10^{-7} \text{ m} = 1237.5 \text{ Å}$$

(iii) Rest mass of a photon is always zero (a photon exists only in motion at speed c).

$$m_{\text{dynamic}} = \frac{E}{c^2} = \frac{1.6 \times 10^{-18}}{(3 \times 10^8)^2} = 1.78 \times 10^{-35} \text{ kg}$$

Final Answer

$$p = 5.33 \times 10^{-27} \text{ kg m/s}, \quad \lambda = 1237.5 \text{ Å}, \quad \text{rest mass} = 0, \quad \text{dynamic mass} = 1.78 \times 10^{-35} \text{ kg}$$

Common Mistake: A photon's *rest* mass is zero; only its dynamic (relativistic) mass E/c^2 is non-zero.

Question 7

Calculate the threshold frequency for photoelectric emission from a metal whose work function is 0.1 eV.

Given: $W_0 = 0.1 \text{ eV} = 1.6 \times 10^{-20} \text{ J}$

Formula: $\nu_0 = \frac{W_0}{h}$

Solution:

$$\nu_0 = \frac{1.6 \times 10^{-20}}{6.6 \times 10^{-34}} = 2.42 \times 10^{13} \text{ Hz}$$

Final Answer

$$\nu_0 \approx 2.4 \times 10^{13} \text{ Hz}$$

Question 8

The work function of sodium is 2.3 eV. Calculate the maximum wavelength (in nm) of light that can cause photoelectric emission from sodium. (Take $h = 6.63 \times 10^{-34} \text{ Js}$)

Given: $W_0 = 2.3 \text{ eV} = 3.68 \times 10^{-19} \text{ J}$, $h = 6.63 \times 10^{-34} \text{ Js}$

Formula: $\lambda_0 = \frac{hc}{W_0}$

Solution:

$$\lambda_0 = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{3.68 \times 10^{-19}} = 5.405 \times 10^{-7} \text{ m} = 540.5 \text{ nm}$$

Final Answer

$$\lambda_0 \approx 540.5 \text{ nm}$$

Common Mistake: This is the *maximum* (threshold) wavelength – any wavelength longer than this will *not* cause emission.

Question 9

Photons of wavelength $\lambda_1 = 5.0 \times 10^{-5} \text{ cm}$ have a total energy equal to the energy of one photon of wavelength $\lambda_2 = 2.0 \times 10^{-12} \text{ cm}$. Find the number of λ_1 -photons, and identify the region of the electromagnetic spectrum to which λ_1 and λ_2 belong.

Given: $\lambda_1 = 5.0 \times 10^{-5} \text{ cm} = 500 \text{ nm}$, $\lambda_2 = 2.0 \times 10^{-12} \text{ cm} = 2.0 \times 10^{-14} \text{ m}$

Formula: $n_1 \left(\frac{hc}{\lambda_1} \right) = \frac{hc}{\lambda_2} \Rightarrow n_1 = \frac{\lambda_1}{\lambda_2}$

Solution:

$$n_1 = \frac{5.0 \times 10^{-7}}{2.0 \times 10^{-14}} = 2.5 \times 10^7$$

$\lambda_1 = 500 \text{ nm}$ lies in the **visible** region; $\lambda_2 = 2.0 \times 10^{-14} \text{ m}$ is an extremely short wavelength lying in the **gamma-ray** region.

Final Answer

$n_1 = 2.5 \times 10^7$ photons; λ_1 : visible light, λ_2 : gamma rays

Common Mistake: Convert both wavelengths from cm to metres (or keep both in the same unit) before taking the ratio.

Question 10

5% of the energy of a 100 W monochromatic source is converted into light of wavelength 560 nm. Find the number of photons emitted per second.

Given: $P_{\text{light}} = 5\% \text{ of } 100 \text{ W} = 5 \text{ W}$, $\lambda = 560 \text{ nm}$

Formula: $E_{\text{photon}} = \frac{hc}{\lambda}$, $n = \frac{P_{\text{light}}}{E_{\text{photon}}}$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{560 \times 10^{-9}} = 3.536 \times 10^{-19} \text{ J}$$

$$n = \frac{5}{3.536 \times 10^{-19}} = 1.414 \times 10^{19} \text{ photons/s}$$

Final Answer

$n \approx 1.41 \times 10^{19}$ photons per second

Question 11

The work function of a metal is 4.2 eV. Will radiation of wavelength 330 nm incident on the surface emit photoelectrons? Justify your answer by calculation.

Given: $W_0 = 4.2 \text{ eV}$, $\lambda = 330 \text{ nm}$

Formula: $\lambda_0 = \frac{hc}{W_0}$

Solution:

$$\lambda_0 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4.2 \times 1.6 \times 10^{-19}} = \frac{1.98 \times 10^{-25}}{6.72 \times 10^{-19}} = 2.946 \times 10^{-7} \text{ m} = 294.6 \text{ nm}$$

Since the incident wavelength (330 nm) is *greater* than the threshold wavelength (294.6 nm), the photon energy is less than the work function.

Final Answer

No photoelectrons will be emitted, since $\lambda_{\text{incident}} > \lambda_0$

Common Mistake: Compare wavelengths correctly: emission occurs only when $\lambda_{\text{incident}} \leq \lambda_0$ (i.e. photon energy $\geq W_0$), not the reverse.

Question 12

A photon ejects an electron with negligible kinetic energy from a metal surface whose threshold frequency is 4.6×10^{14} Hz. Calculate the energy of the incident photon, in eV.

Given: $\nu_0 = 4.6 \times 10^{14}$ Hz, negligible KE $\Rightarrow$ incident photon energy $= W_0 = h\nu_0$

Formula: $E = h\nu_0$

Solution:

$$E = 6.6 \times 10^{-34} \times 4.6 \times 10^{14} = 3.036 \times 10^{-19} \text{ J} = \frac{3.036 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.8975 \text{ eV}$$

Final Answer

$E \approx 1.9 \text{ eV}$

II. Einstein's Photoelectric Equation

Question 13

Assuming the visible spectrum extends from 400 nm to 750 nm, calculate the maximum work function (in eV) of a photocathode that can emit photoelectrons using visible light of any wavelength in this range.

Given: Visible range: 400 nm to 750 nm

Formula: $W_{0(\max)} = \frac{hc}{\lambda_{\max}}$

Solution:

For emission to occur at *any* wavelength in the visible range (including the least energetic, longest wavelength of 750 nm), the work function must not exceed the photon energy at 750 nm.

$$W_{0(\max)} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{750 \times 10^{-9}} = 2.64 \times 10^{-19} \text{ J} = \frac{2.64 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.65 \text{ eV}$$

Final Answer

$$W_{0(\max)} \approx 1.65 \text{ eV}$$

Common Mistake: Use the *longest* wavelength (lowest photon energy) of the range, not the shortest, since the work function must be small enough for even the least energetic photon to eject an electron.

Question 14

A metal sheet carries a negative charge of 11.2 nC. Find the number of ultraviolet photons needed to completely discharge the sheet by photoelectron ejection, and the minimum energy that must be absorbed by the sheet to effect this discharge. The threshold frequency of the metal is 4.5×10^{15} Hz.

Given: $Q = 11.2 \text{ nC} = 11.2 \times 10^{-9} \text{ C}$, $\nu_0 = 4.5 \times 10^{15} \text{ Hz}$

Formula: $n = \frac{Q}{e}$, $W_{\min} = n W_0 = n h \nu_0$

Solution:

Each photon ejects (at most) one electron, so the number of photons needed equals the

number of electrons to be removed:

$$n = \frac{11.2 \times 10^{-9}}{1.6 \times 10^{-19}} = 7.0 \times 10^{10}$$

$$W_0 = h\nu_0 = 6.6 \times 10^{-34} \times 4.5 \times 10^{15} = 2.97 \times 10^{-18} \text{ J}$$

$$W_{\min} = n W_0 = 7.0 \times 10^{10} \times 2.97 \times 10^{-18} = 2.08 \times 10^{-7} \text{ J}$$

Final Answer

Number of photons = 7.0×10^{10} ; minimum energy $\approx 2.08 \times 10^{-7} \text{ J}$

Question 15

The work function of copper is 4 eV. Monochromatic light of wavelength 4000 Å falls on it. Calculate the energy of the incident photon in joule, express the work function in joule, and find the kinetic energy of the fastest ejected electron.

★ Likely Exam Question

Given: $W_0 = 4 \text{ eV}$, $\lambda = 4000 \text{ Å} = 4 \times 10^{-7} \text{ m}$

Formula: $E_{\text{photon}} = \frac{hc}{\lambda}$, $E_{k(\max)} = E_{\text{photon}} - W_0$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4 \times 10^{-7}} = 4.95 \times 10^{-19} \text{ J}$$

$$W_0 = 4 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-19} \text{ J}$$

Since $E_{\text{photon}} (4.95 \times 10^{-19} \text{ J}) < W_0 (6.4 \times 10^{-19} \text{ J})$, the photon does not carry enough energy to eject an electron.

Final Answer

$E_{\text{photon}} = 4.95 \times 10^{-19} \text{ J}$, $W_0 = 6.4 \times 10^{-19} \text{ J}$ – no electrons will be ejected

Common Mistake: Do not blindly subtract to get a "negative kinetic energy" – recognise that KE cannot be negative, so no emission occurs.

Question 16

Calculate the work function of sodium metal in eV, given that the threshold wavelength for photoelectric emission is 5400 Å. Will photoelectrons be emitted if red light of wavelength 6800 Å is incident on sodium?

Given: $\lambda_0 = 5400 \text{ Å} = 5.4 \times 10^{-7} \text{ m}$, $\lambda = 6800 \text{ Å}$

Formula: $W_0 = \frac{hc}{\lambda_0}$

Solution:

$$W_0 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5.4 \times 10^{-7}} = 3.667 \times 10^{-19} \text{ J} = \frac{3.667 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.29 \text{ eV}$$

Since $6800 \text{ \AA} > 5400 \text{ \AA}$ (threshold), the red light has lower photon energy than W_0 .

Final Answer

$W_0 \approx 2.3 \text{ eV}$; **No**, red light of $6800 \text{ \AA}$ cannot emit photoelectrons from sodium

Common Mistake: Compare the given wavelength directly with the threshold wavelength – longer wavelength means lower energy, hence no emission.

Question 17

Electromagnetic radiation of frequency $3 \times 10^{15} \text{ Hz}$ falls on a photoelectric surface whose work function is 4.0 eV . Find the maximum velocity of the emitted photoelectrons.

★ Likely Exam Question

Given: $\nu = 3 \times 10^{15} \text{ Hz}$, $W_0 = 4.0 \text{ eV} = 6.4 \times 10^{-19} \text{ J}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$

Formula: $E_{k(\max)} = h\nu - W_0$, $v_{\max} = \sqrt{\frac{2E_{k(\max)}}{m_e}}$

Solution:

$$E_{k(\max)} = 6.6 \times 10^{-34} \times 3 \times 10^{15} - 6.4 \times 10^{-19} = 1.98 \times 10^{-18} - 6.4 \times 10^{-19} = 1.34 \times 10^{-18} \text{ J}$$

$$v_{\max} = \sqrt{\frac{2 \times 1.34 \times 10^{-18}}{9.1 \times 10^{-31}}} = \sqrt{2.945 \times 10^{12}} = 1.716 \times 10^6 \text{ m/s}$$

Final Answer

$v_{\max} \approx 1.72 \times 10^6 \text{ m/s}$

Question 18

The maximum kinetic energy of photoelectrons is $10 \times 10^{-19} \text{ J}$ when a surface is illuminated by light of wavelength 100 nm . Find the work function of the surface.

Given: $E_{k(\max)} = 10 \times 10^{-19} \text{ J}$, $\lambda = 100 \text{ nm}$

Formula: $W_0 = \frac{hc}{\lambda} - E_{k(\max)}$

Solution:

$$E_{\text{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{100 \times 10^{-9}} = 1.98 \times 10^{-18} \text{ J}$$

$$W_0 = 1.98 \times 10^{-18} - 1.0 \times 10^{-18} = 0.98 \times 10^{-18} \text{ J} = \frac{0.98 \times 10^{-18}}{1.6 \times 10^{-19}} = 6.125 \text{ eV}$$

Final Answer

$$W_0 \approx 6.13 \text{ eV}$$

III. Stopping Potential

Question 19

A metal whose work function is 3.0 eV is illuminated by light of wavelength 3×10^{-7} m. Calculate (i) the threshold frequency, (ii) the maximum kinetic energy of the photoelectrons, and (iii) the stopping potential.

★ Likely Exam Question

Given: $W_0 = 3.0 \text{ eV} = 4.8 \times 10^{-19} \text{ J}$, $\lambda = 3 \times 10^{-7} \text{ m}$

Formula: $\nu_0 = \frac{W_0}{h}$, $E_{k(\max)} = h\nu - W_0$, $eV_0 = E_{k(\max)}$

Solution:

(i)

$$\nu_0 = \frac{4.8 \times 10^{-19}}{6.6 \times 10^{-34}} = 7.27 \times 10^{14} \text{ Hz}$$

(ii)

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{3 \times 10^{-7}} = 1 \times 10^{15} \text{ Hz}$$

$$E_{k(\max)} = h\nu - W_0 = 6.6 \times 10^{-19} - 4.8 \times 10^{-19} = 1.8 \times 10^{-19} \text{ J}$$

(iii)

$$V_0 = \frac{E_{k(\max)}}{e} = \frac{1.8 \times 10^{-19}}{1.6 \times 10^{-19}} = 1.125 \text{ V}$$

Final Answer

$$\nu_0 \approx 0.73 \times 10^{15} \text{ Hz}, \quad E_{k(\max)} = 1.8 \times 10^{-19} \text{ J}, \quad V_0 = 1.125 \text{ V}$$

Question 20

When light of wavelength 300 nm is incident on a sodium cathode, the stopping potential is 1.85 V; when the wavelength is 400 nm, the stopping potential becomes 0.82 V. Calculate (i) Planck's constant, the work function of sodium, and the threshold wavelength for sodium. (ii) If the wavelength is kept fixed at 400 nm but the intensity of the light is doubled, will the photoelectric current increase? Give reasons.

★ Likely Exam Question

Given: $\lambda_1 = 300 \text{ nm}$, $V_{01} = 1.85 \text{ V}$; $\lambda_2 = 400 \text{ nm}$, $V_{02} = 0.82 \text{ V}$

Formula: $eV_0 = \frac{hc}{\lambda} - W_0$

Solution:

Writing Einstein's equation for both cases and subtracting:

$$e(V_{01} - V_{02}) = hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$$

$$1.6 \times 10^{-19} \times 1.03 = hc \left(\frac{1}{3 \times 10^{-7}} - \frac{1}{4 \times 10^{-7}} \right) = hc \times 8.333 \times 10^5$$

$$hc = \frac{1.648 \times 10^{-19}}{8.333 \times 10^5} = 1.978 \times 10^{-25} \Rightarrow h = \frac{1.978 \times 10^{-25}}{3 \times 10^8} = 6.6 \times 10^{-34} \text{ Js}$$

Work function:

$$W_0 = \frac{hc}{\lambda_1} - eV_{01} = 6.59 \times 10^{-19} - 2.96 \times 10^{-19} = 3.63 \times 10^{-19} \text{ J} = 2.27 \text{ eV}$$

Threshold wavelength:

$$\lambda_0 = \frac{hc}{W_0} = \frac{1.978 \times 10^{-25}}{3.63 \times 10^{-19}} = 5.45 \times 10^{-7} \text{ m} = 5450 \text{ Å}$$

(ii) The stopping potential depends only on the photon energy (frequency), not on the intensity of light. At a fixed wavelength, doubling the intensity only increases the *number* of photons striking the surface per second – so more photoelectrons are ejected per second and the **saturation current increases**, but the **stopping potential remains unchanged**, since the maximum KE of individual electrons depends only on ν .

Final Answer

$h \approx 6.6 \times 10^{-34} \text{ Js}$, $W_0 \approx 2.27 \text{ eV}$, $\lambda_0 \approx 5450 \text{ Å}$; photoelectric current *does* increase with intensity (more electrons/s), but stopping potential stays the same

Common Mistake: Do not confuse "stopping potential unchanged" with "current unchanged" – intensity affects the number of photoelectrons (current), not their maximum KE (stopping potential).

Question 21

A retarding potential of 0.5 V is required to stop the photoelectrons emitted from the cathode of a photoelectric cell when violet light of wavelength 400 nm strikes its surface. Find (i) the work function of the surface, (ii) the threshold frequency, (iii) the energy of light of wavelength 300 nm, and (iv) the maximum kinetic energy of electrons emitted when 300 nm light strikes the surface.

★ **Likely Exam Question**

Given: $V_0 = 0.5 \text{ V}$, $\lambda = 400 \text{ nm}$

Formula: $W_0 = \frac{hc}{\lambda} - eV_0, \quad \nu_0 = \frac{W_0}{h}, \quad E_{k(\max)} = E'_{\text{photon}} - W_0$

Solution:

(i)

$$\frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{400 \times 10^{-9}} = 4.95 \times 10^{-19} \text{ J}$$

$$eV_0 = 0.5 \times 1.6 \times 10^{-19} = 0.8 \times 10^{-19} \text{ J}$$

$$W_0 = 4.95 \times 10^{-19} - 0.8 \times 10^{-19} = 4.15 \times 10^{-19} \text{ J}$$

(ii)

$$\nu_0 = \frac{4.15 \times 10^{-19}}{6.6 \times 10^{-34}} = 6.29 \times 10^{14} \text{ Hz}$$

(iii)

$$E = \frac{hc}{300 \times 10^{-9}} = \frac{1.98 \times 10^{-25}}{3 \times 10^{-7}} = 6.6 \times 10^{-19} \text{ J}$$

(iv)

$$E_{k(\max)} = 6.6 \times 10^{-19} - 4.15 \times 10^{-19} = 2.45 \times 10^{-19} \text{ J}$$

Final Answer

$$W_0 = 4.15 \times 10^{-19} \text{ J}, \quad \nu_0 \approx 6.3 \times 10^{14} \text{ Hz}, \quad E = 6.6 \times 10^{-19} \text{ J}, \quad E_{k(\max)} = 2.45 \times 10^{-19} \text{ J}$$

Question 22

Ultraviolet light of wavelengths 80 nm and 70 nm, when allowed to fall on hydrogen atoms in their ground state, liberates electrons with kinetic energies 1.8 eV and 4.0 eV respectively. Find the value of Planck's constant.

Given: $\lambda_1 = 80 \text{ nm}, E_{k1} = 1.8 \text{ eV}; \quad \lambda_2 = 70 \text{ nm}, E_{k2} = 4.0 \text{ eV}$

Formula: $E_{k1} = \frac{hc}{\lambda_1} - W, \quad E_{k2} = \frac{hc}{\lambda_2} - W$

Solution:

Subtracting the two equations to eliminate the (ionization) work W :

$$E_{k2} - E_{k1} = hc \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

$$(4.0 - 1.8) \times 1.6 \times 10^{-19} = hc \left(\frac{1}{70 \times 10^{-9}} - \frac{1}{80 \times 10^{-9}} \right)$$

$$3.52 \times 10^{-19} = hc \times 1.786 \times 10^6$$

$$hc = 1.971 \times 10^{-25} \Rightarrow h = \frac{1.971 \times 10^{-25}}{3 \times 10^8} = 6.57 \times 10^{-34} \text{ Js}$$

Final Answer

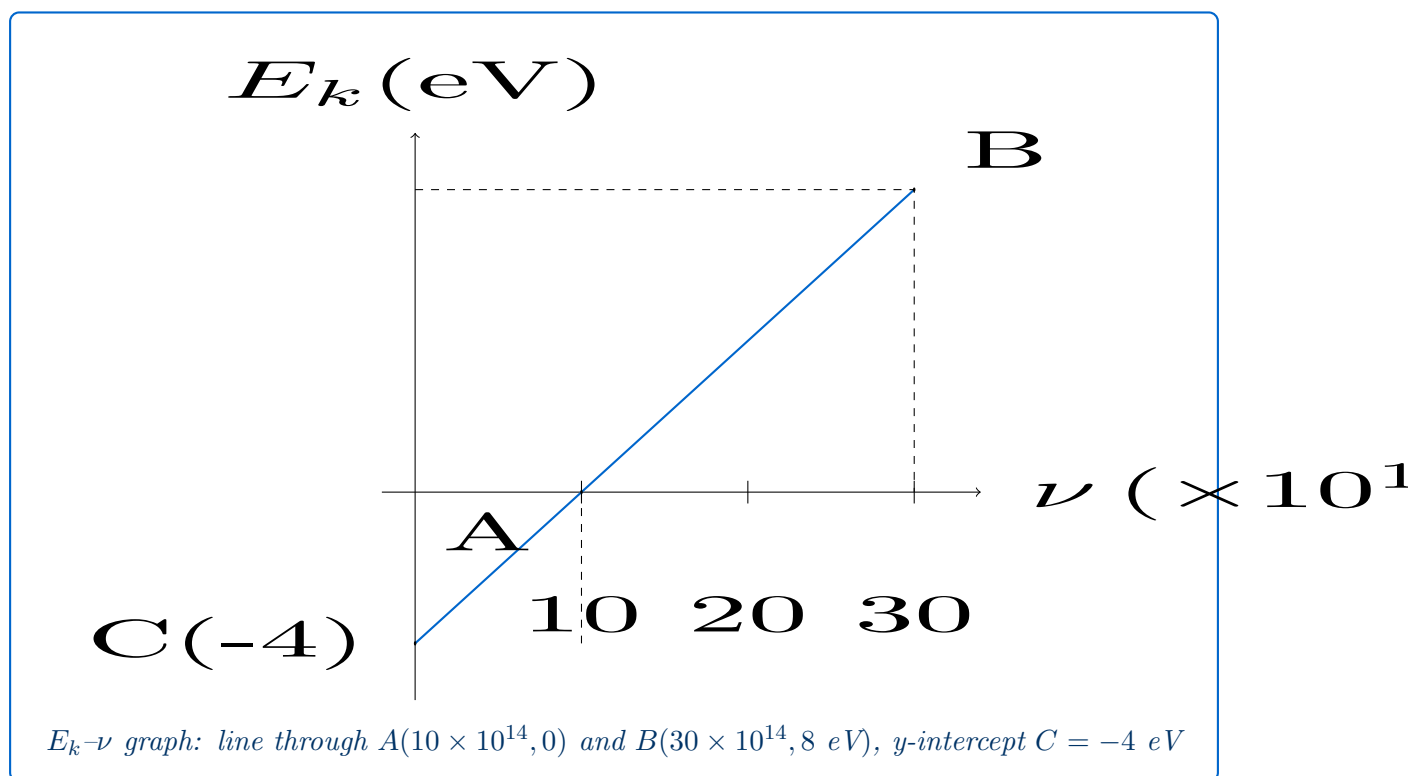
$$h \approx 6.57 \times 10^{-34} \text{ Js}$$

Common Mistake: Set up two simultaneous Einstein equations and subtract to eliminate the unknown work term – do not try to solve for h from a single equation.

Question 23

In an experiment on the photoelectric effect, the graph of the maximum kinetic energy E_k of the emitted photoelectrons versus the frequency ν of the incident light is a straight line, as shown in the figure. Calculate (i) the threshold frequency, (ii) the work function W of the cathode metal (in eV), and (iii) the stopping potential for photoelectrons emitted by light of frequency $\nu = 30 \times 10^{14} \text{ Hz}$.

★ Likely Exam Question



Given: From graph: line passes through $A(10 \times 10^{14} \text{ Hz}, 0)$ and $B(30 \times 10^{14} \text{ Hz}, 8 \text{ eV})$; y-intercept at -4 eV

Formula: $E_k = h\nu - W$ (equation of a straight line: x-intercept $= \nu_0$, y-intercept $= -W$)

Solution:

(i) The line cuts the ν -axis (where $E_k = 0$) at $\nu = 10 \times 10^{14} \text{ Hz}$.

$$\nu_0 = 10 \times 10^{14} \text{ Hz} = 1 \times 10^{15} \text{ Hz}$$

(ii) At $\nu = 0$ (y-intercept), $E_k = -W$. From the graph this intercept is -4 eV.

$$W = 4 \text{ eV}$$

(iii) At $\nu = 30 \times 10^{14}$ Hz, the graph gives $E_k(\text{max}) = 8$ eV. Since $eV_0 = E_k(\text{max})$,

$$V_0 = 8 \text{ V}$$

Final Answer

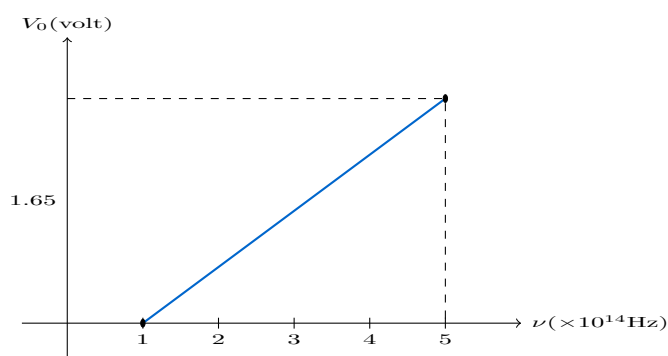
$$\nu_0 = 10 \times 10^{14} \text{ Hz}, \quad W = 4 \text{ eV}, \quad V_0 = 8 \text{ V}$$

Common Mistake: Read the y-intercept (at $\nu = 0$) as $-W$, not $+W$ – the line's equation is $E_k = h\nu - W$.

Question 24

The given figure shows the plot of stopping potential V_0 versus the frequency ν of light used in a photoelectric-effect experiment. Calculate the value of Planck's constant h . (Take $e = 1.6 \times 10^{-19}$ C)

★ Likely Exam Question



V_0 - ν graph: line through $(1 \times 10^{14}, 0)$ and $(5 \times 10^{14}, 1.65 \text{ V})$

Given: From graph: line passes through $(1 \times 10^{14} \text{ Hz}, 0)$ and $(5 \times 10^{14} \text{ Hz}, 1.65 \text{ V})$

Formula: $eV_0 = h\nu - W_0 \Rightarrow \text{slope of } V_0\text{-}\nu \text{ graph} = \frac{h}{e}$

Solution:

$$\text{slope} = \frac{1.65 - 0}{(5 - 1) \times 10^{14}} = \frac{1.65}{4 \times 10^{14}} = 4.125 \times 10^{-15} \text{ V s}$$

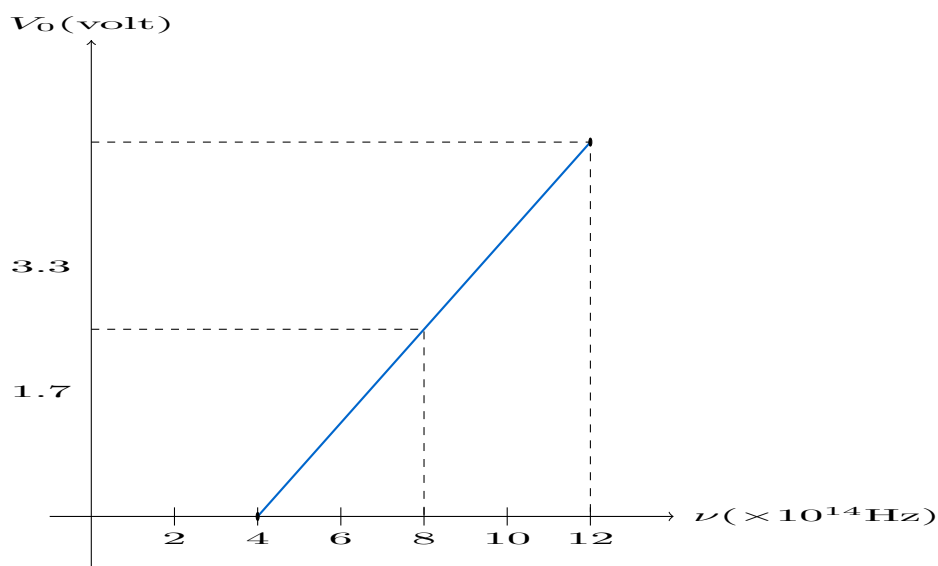
$$h = e \times \text{slope} = 1.6 \times 10^{-19} \times 4.125 \times 10^{-15} = 6.6 \times 10^{-34} \text{ J s}$$

Final Answer

$$h \approx 6.6 \times 10^{-34} \text{ J s}$$

Question 25

The stopping potential V_0 (in volt) as a function of frequency ν for a sodium emitter is shown in the figure. Using only the data plotted on the graph – i.e. without assuming the value of Planck’s constant beforehand – find the work function of sodium.

★ Likely Exam Question

V_0 - ν graph: line through $(4 \times 10^{14}, 0)$ and $(12 \times 10^{14}, 3.3 \text{ V})$

Given: From graph: line passes through $(4 \times 10^{14} \text{ Hz}, 0)$ and $(12 \times 10^{14} \text{ Hz}, 3.3 \text{ V})$

Formula: $W_0 = h\nu_0$, $h = e \times \text{slope of the graph}$

Solution:

Slope of the graph (using only the two plotted points – this gives h/e directly from the graph, without assuming h):

$$\text{slope} = \frac{h}{e} = \frac{3.3 - 0}{(12 - 4) \times 10^{14}} = 4.125 \times 10^{-15} \text{ V s}$$

$$h = e \times 4.125 \times 10^{-15} = 1.6 \times 10^{-19} \times 4.125 \times 10^{-15} = 6.6 \times 10^{-34} \text{ J s}$$

The threshold frequency (x-intercept) read directly from the graph is $\nu_0 = 4 \times 10^{14} \text{ Hz}$. Hence:

$$W_0 = h\nu_0 = 6.6 \times 10^{-34} \times 4 \times 10^{14} = 2.64 \times 10^{-19} \text{ J}$$

Final Answer

$$W_0 = 2.64 \times 10^{-19} \text{ J } (\approx 1.65 \text{ eV})$$

Common Mistake: "Without using the known value of h " means *derive* h from the slope of the graph itself, rather than quoting a memorised textbook value.

Question 26

When a certain metallic surface is illuminated with monochromatic light of wavelength λ , the stopping potential for the photoelectric current is $3V_0$. When the same surface is illuminated with light of wavelength 2λ , the stopping potential is V_0 . Find the threshold wavelength for the surface.

Given: At λ : stopping potential = $3V_0$; at 2λ : stopping potential = V_0

Formula: $eV'_0 = \frac{hc}{\lambda'} - W_0$

Solution:

$$e(3V_0) = \frac{hc}{\lambda} - W_0 \quad \dots(1)$$

$$e(V_0) = \frac{hc}{2\lambda} - W_0 \quad \dots(2)$$

Subtracting (2) from (1):

$$2eV_0 = \frac{hc}{\lambda} - \frac{hc}{2\lambda} = \frac{hc}{2\lambda} \Rightarrow eV_0 = \frac{hc}{4\lambda}$$

Substituting back into (2):

$$W_0 = \frac{hc}{2\lambda} - eV_0 = \frac{hc}{2\lambda} - \frac{hc}{4\lambda} = \frac{hc}{4\lambda}$$

Threshold wavelength:

$$\lambda_0 = \frac{hc}{W_0} = \frac{hc}{\left(\frac{hc}{4\lambda}\right)} = 4\lambda$$

Final Answer

$$\lambda_0 = 4\lambda$$

Common Mistake: Keep λ symbolic throughout – do not substitute a numeric value where none is given; the answer is meant to be expressed in terms of λ .

Question 27

When the energy of the incident radiation is increased by 40%, the kinetic energy of the photoelectrons emitted from a metal surface increases from 0.5 eV to 1.0 eV. Find the work function of the metal surface.

Given: $E_{k1} = 0.5 \text{ eV}$, $E_{k2} = 1.0 \text{ eV}$, $E_2 = 1.4E_1$

Formula: $E_{k1} = E_1 - W_0$, $E_{k2} = E_2 - W_0$

Solution:

Subtracting the equations:

$$E_{k2} - E_{k1} = E_2 - E_1 = 1.4E_1 - E_1 = 0.4E_1$$

$$0.4E_1 = 1.0 - 0.5 = 0.5 \text{ eV} \Rightarrow E_1 = 1.25 \text{ eV}$$

$$W_0 = E_1 - E_{k1} = 1.25 - 0.5 = 0.75 \text{ eV}$$

Final Answer

$$W_0 = 0.75 \text{ eV}$$

Question 28

In a photoelectric experiment, ultraviolet light of wavelength 280 nm is used with a lithium cathode of work function $\phi = 2.5 \text{ eV}$. If the wavelength of the incident light is switched to 400 nm, find the change in stopping potential.

Given: $\lambda_1 = 280 \text{ nm}$, $\lambda_2 = 400 \text{ nm}$, $\phi = 2.5 \text{ eV}$

Formula: $eV_0 = \frac{hc}{\lambda} - \phi \quad \left(\frac{hc}{e} \approx 1237.5 \text{ eV nm} \right)$

Solution:

$$\frac{hc}{e\lambda_1} = \frac{1237.5}{280} = 4.42 \text{ eV} \Rightarrow V_{01} = 4.42 - 2.5 = 1.92 \text{ V}$$

$$\frac{hc}{e\lambda_2} = \frac{1237.5}{400} = 3.09 \text{ eV} \Rightarrow V_{02} = 3.09 - 2.5 = 0.59 \text{ V}$$

$$\Delta V_0 = V_{01} - V_{02} = 1.92 - 0.59 = 1.33 \text{ V}$$

Final Answer

Stopping potential decreases by about 1.3 V on switching from 280 nm to 400 nm

IV. de Broglie Wavelength When Momentum or Velocity is Given

Question 29

What is the de Broglie wavelength of a 2 kg object moving at a speed of 1 m/s?

Given: $m = 2 \text{ kg}$, $v = 1 \text{ m/s}$

Formula: $\lambda = \frac{h}{mv}$

Solution:

$$\lambda = \frac{6.6 \times 10^{-34}}{2 \times 1} = 3.3 \times 10^{-34} \text{ m}$$

Final Answer

$$\lambda = 3.3 \times 10^{-34} \text{ m}$$

Common Mistake: This confirms why de Broglie waves are never observed for macroscopic objects – the wavelength is immeasurably small.

Question 30

Calculate the de Broglie wavelength of a proton whose momentum is $2.55 \times 10^{-22} \text{ kg m/s}$.

Given: $p = 2.55 \times 10^{-22} \text{ kg m/s}$

Formula: $\lambda = \frac{h}{p}$

Solution:

$$\lambda = \frac{6.6 \times 10^{-34}}{2.55 \times 10^{-22}} = 2.588 \times 10^{-12} \text{ m}$$

Final Answer

$$\lambda \approx 2.59 \times 10^{-12} \text{ m}$$

Question 31

A moving particle has a de Broglie wavelength of $1.0 \text{ \AA}$. What is the momentum of the particle?

Given: $\lambda = 1.0 \text{ \AA} = 1 \times 10^{-10} \text{ m}$

Formula: $p = \frac{h}{\lambda}$

Solution:

$$p = \frac{6.6 \times 10^{-34}}{1 \times 10^{-10}} = 6.6 \times 10^{-24} \text{ kg m/s}$$

Final Answer

$$p = 6.6 \times 10^{-24} \text{ kg m/s}$$

Question 32

Calculate the de Broglie wavelength of electrons moving with a speed of 0.5 km/s .

Given: $v = 0.5 \text{ km/s} = 500 \text{ m/s}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$

Formula: $\lambda = \frac{h}{m_e v}$

Solution:

$$\lambda = \frac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \times 500} = \frac{6.6 \times 10^{-34}}{4.55 \times 10^{-28}} = 1.45 \times 10^{-6} \text{ m}$$

Final Answer

$$\lambda \approx 1.45 \times 10^{-6} \text{ m}$$

V. de Broglie Wavelength When Kinetic Energy is Given

Question 33

Find the de Broglie wavelength of a proton of energy 1.0 keV.

Given: $E_k = 1.0 \text{ keV} = 1000 \text{ eV} = 1.6 \times 10^{-16} \text{ J}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$

Formula: $\lambda = \frac{h}{\sqrt{2m_p E_k}}$

Solution:

$$2m_p E_k = 2 \times 1.67 \times 10^{-27} \times 1.6 \times 10^{-16} = 5.344 \times 10^{-43}$$

$$\sqrt{5.344 \times 10^{-43}} = 7.31 \times 10^{-22}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{7.31 \times 10^{-22}} = 9.03 \times 10^{-13} \text{ m} = 9.0 \times 10^{-3} \text{ \AA}$$

Final Answer

$$\lambda \approx 9.0 \times 10^{-13} \text{ m} = 9.0 \times 10^{-3} \text{ \AA}$$

Common Mistake: Convert keV to J via eV (1 keV = 1000 eV) before using it inside the square root – a common slip is forgetting the factor of 1000.

Question 34

Find the de Broglie wavelength of a neutron of mass $1.67 \times 10^{-24} \text{ g}$ moving with kinetic energy 0.04 eV.

Given: $m_n = 1.67 \times 10^{-24} \text{ g} = 1.67 \times 10^{-27} \text{ kg}$, $E_k = 0.04 \text{ eV} = 6.4 \times 10^{-21} \text{ J}$

Formula: $\lambda = \frac{h}{\sqrt{2m_n E_k}}$

Solution:

$$2m_n E_k = 2 \times 1.67 \times 10^{-27} \times 6.4 \times 10^{-21} = 2.138 \times 10^{-47}$$

$$\sqrt{2.138 \times 10^{-47}} = 4.623 \times 10^{-24}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{4.623 \times 10^{-24}} = 1.428 \times 10^{-10} \text{ m} = 1.43 \text{ \AA}$$

Final Answer

$$\lambda \approx 1.43 \text{ \AA}$$

Common Mistake: Convert the mass from grams to kilograms first (divide by 1000)
– an easy unit slip when the mass is given in g.

VI. de Broglie Wavelength When Accelerating Potential is Given

Question 35

Calculate the de Broglie wavelength of an electron of energy 400 eV.

★ Likely Exam Question

Given: $V = 400 \text{ V}$ (equivalently, $E_k = 400 \text{ eV}$)

Formula: $\lambda = \frac{h}{\sqrt{2m_e eV}} \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$

Solution:

$$\lambda = \frac{12.27}{\sqrt{400}} = \frac{12.27}{20} = 0.6135 \text{ \AA}$$

Final Answer

$$\lambda \approx 0.61 \text{ \AA}$$

Question 36

Calculate the de Broglie wavelength of an electron beam accelerated through a potential difference of 60 V.

★ Likely Exam Question

Given: $V = 60 \text{ V}$

Formula: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$

Solution:

$$\lambda = \frac{12.27}{\sqrt{60}} = \frac{12.27}{7.746} = 1.584 \text{ \AA}$$

Final Answer

$$\lambda \approx 1.58 \text{ \AA}$$

Question 37

In a diode valve, the anode has a potential of 100 V relative to the cathode. Find the de Broglie wavelength of electrons on reaching the anode.

Given: $V = 100 \text{ V}$

Formula: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA}$

Solution:

$$\lambda = \frac{12.27}{\sqrt{100}} = \frac{12.27}{10} = 1.227 \text{ \AA}$$

Final Answer

$$\lambda \approx 1.23 \text{ \AA}$$

Question 38

What potential must be applied to an electron microscope to produce electrons of wavelength 1.0 Å?

★ **Likely Exam Question**

Given: $\lambda = 1.0 \text{ \AA}$

Formula: $\lambda \approx \frac{12.27}{\sqrt{V}} \text{ \AA} \Rightarrow V = \left(\frac{12.27}{\lambda} \right)^2$

Solution:

$$V = \left(\frac{12.27}{1.0} \right)^2 = 150.55 \text{ V}$$

Final Answer

$$V \approx 150.6 \text{ V}$$

Common Mistake: The formula $\lambda \approx 12.27/\sqrt{V} \text{ \AA}$ applies only to electrons – for protons/neutrons use $\lambda = h/\sqrt{2mE_k}$ directly with the correct mass.